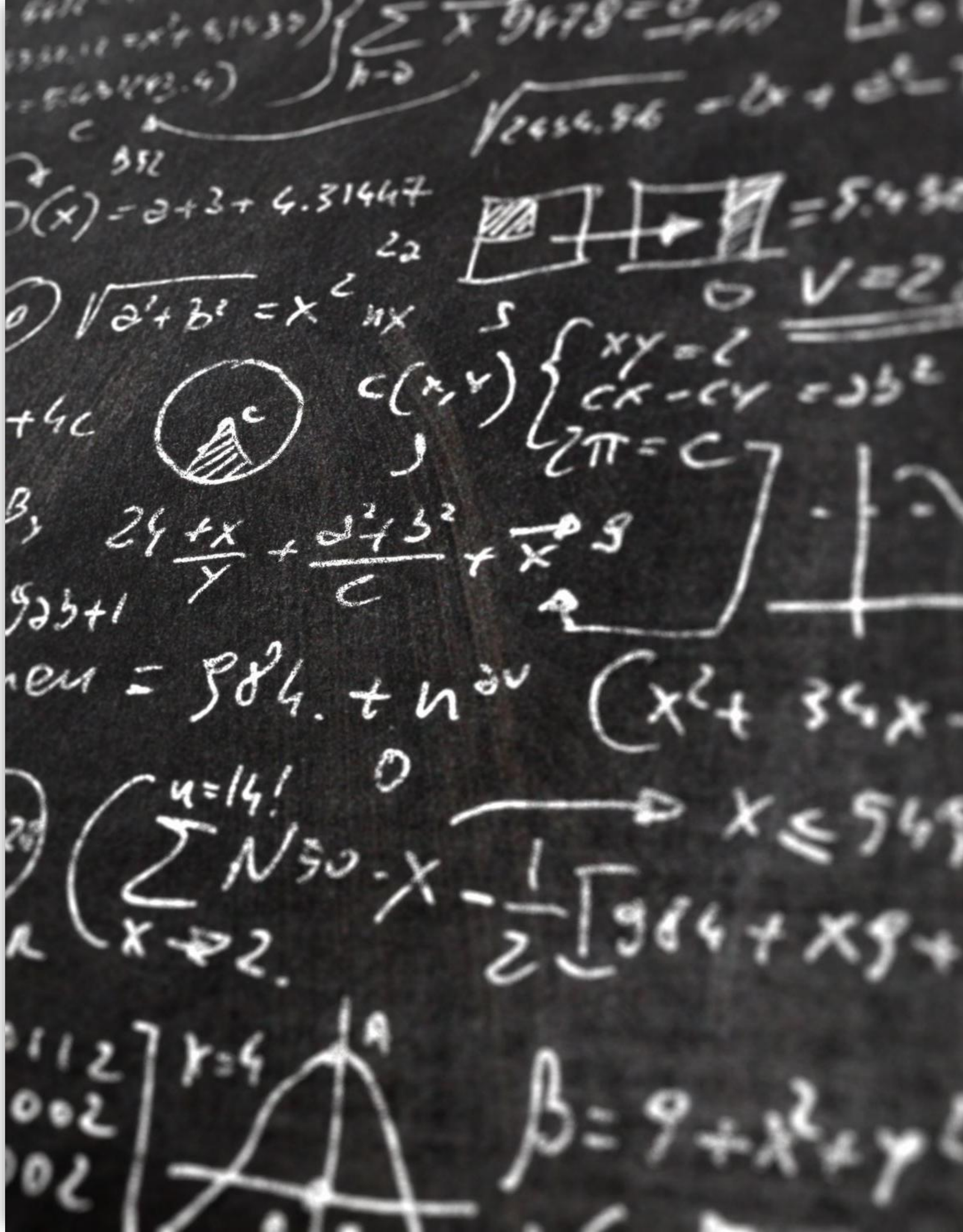
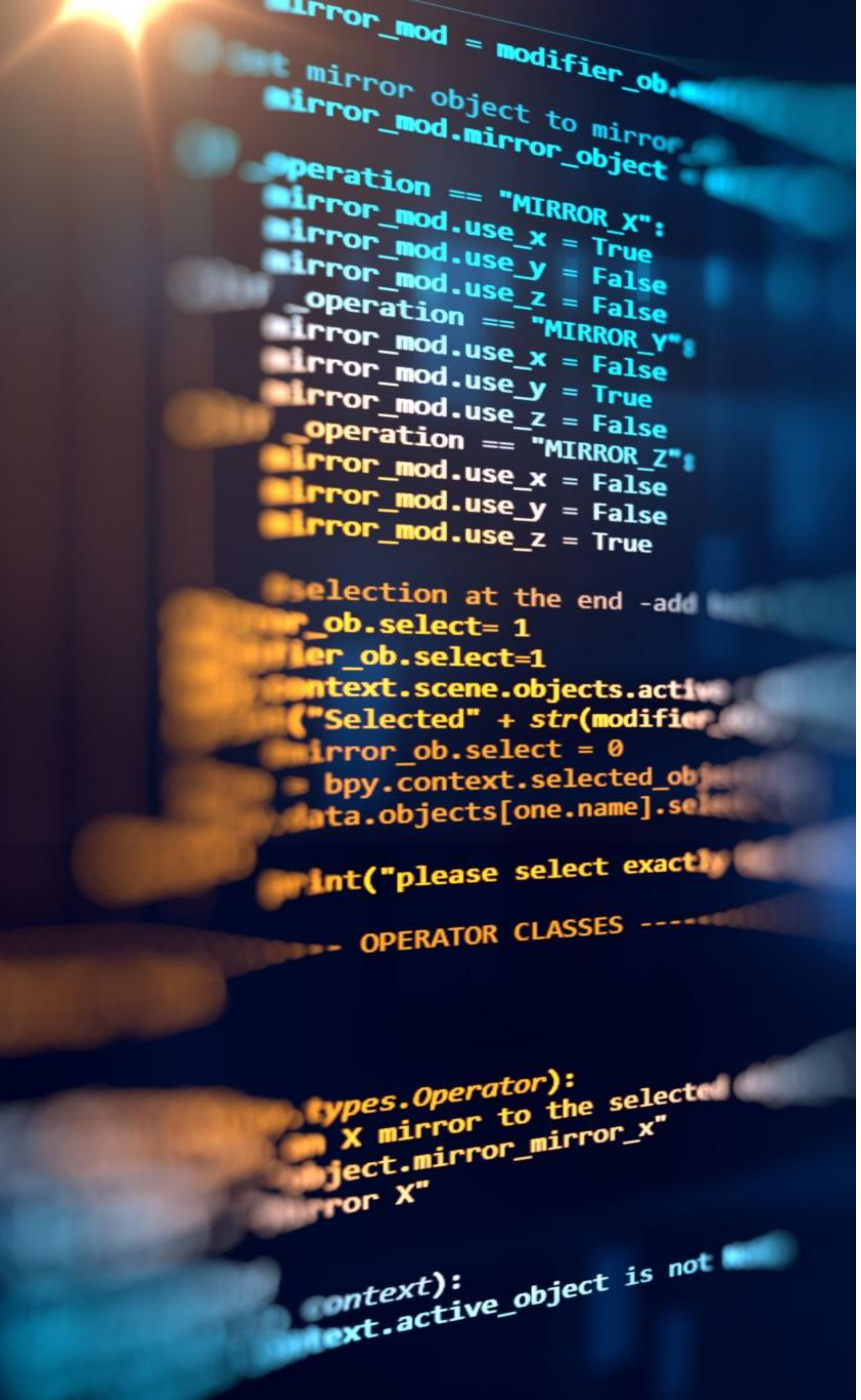
The Fundamentals of Data Modeling

Objectives

- Explain what data modeling is.
- Understand the role of ERD in database designing.
- Identify the components of ER diagram and its role in database designing.
- Explain the process of creating ER diagrams.





What is data modeling?

- **Data modeling** is a process that you use to define the data structure of a database.
- It is a technique that you can use to create a database from scratch.
- *Data modeling is the process of transforming data into information.*

Why data modeling?

- **Data modeling** helps in translating the requirements of business users into a data model that can be used to support business processes and scale analytics.

Why data modeling?

- A good data model should be able to answer all of these questions:
- What are our business processes?
- How do we structure our business information?
- What kinds of information do we use within these processes?
- What kinds of information do we store?
- Where does it come from? Where does it go?

The importance of data modeling

- Data modeling is an important stage of any software project because, without it, you cannot get a clear idea of what your database should look like and how your application will be built upon it.
- A comprehensive and optimized data model helps create a simplified, logical database that eliminates redundancy, reduces storage requirements, and enables efficient retrieval.

Basics of Data Modeling

- A good database design starts with a list of the data that you want to include in your database.
- Best way to represent data modeling is by creating an ER Diagram.

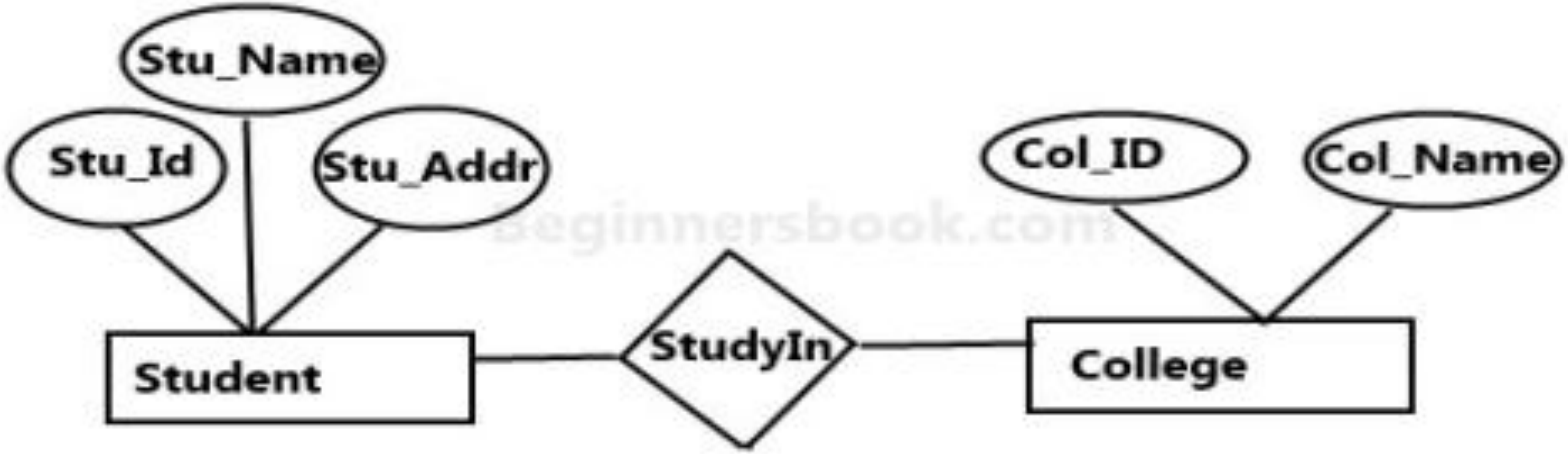
What is an Entity Relationship Diagram?

- An Entity Relationship Diagram is a diagram that represents relationships among entities in a database.
- It is commonly known as an ER Diagram.
- An ER model is a design or blueprint of a database that can later be implemented as a database.

A simple ER Diagram

- In the following diagram we have two entities Student and College and their relationship.
- The relationship between Student and College is many to one as a college can have many students however a student cannot study in multiple colleges at the same time.
- Student entity has attributes such as Stu_Id, Stu_Name & Stu_Addr and College entity has attributes such as Col_ID & Col_Name.

A simple ER Diagram



Sample E-R Diagram

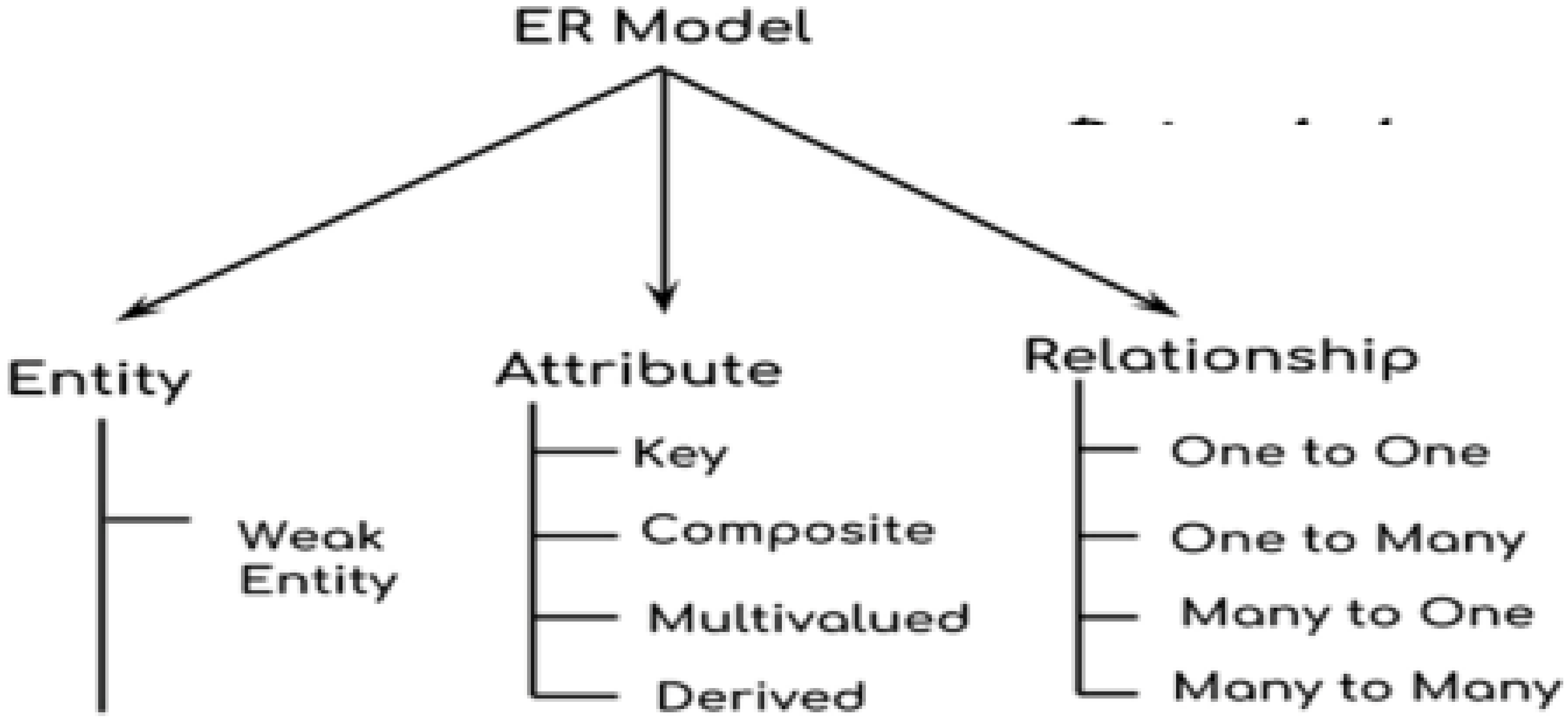
Why use ER Diagrams?

- Provide a preview of how all your tables should connect, what fields are going to be on each table
- Helps to describe entities, attributes, relationships
- ER diagrams are translatable into relational tables which allows you to build databases quickly
- ER diagrams can be used by database designers as a blueprint for implementing data in specific software applications

Shapes and their meaning in an ER Diagram

- Rectangle: Represents Entity sets.
- Ellipses: Attributes
- Diamonds: Relationship Set
- Lines: They link attributes to Entity Sets and Entity sets to Relationship Set
- Double Ellipses: Multivalued Attributes
- Dashed Ellipses: Derived Attributes
- Double Rectangles: Weak Entity Sets
- Double Lines: Total participation of an entity in a relationship set

Components of ER Diagram



Entity

An object tracked in the Database.

Attributes

Properties or traits.

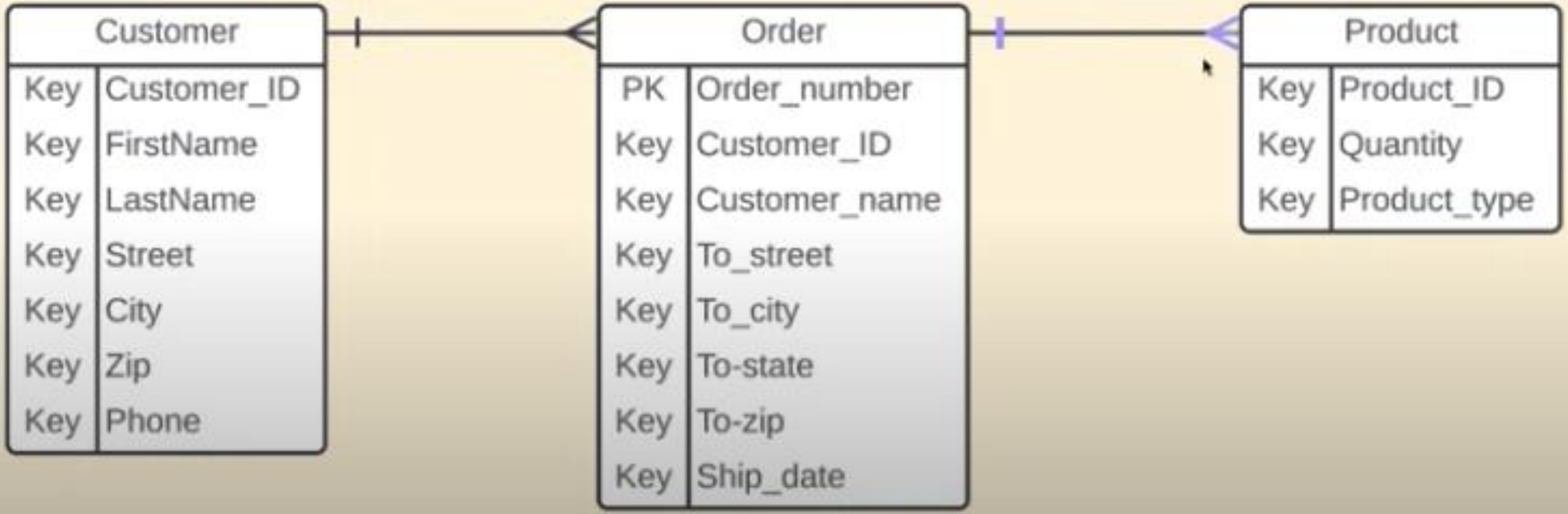
Customer	
Key	Customer_ID
Key	FirstName
Key	LastName
Key	Street
Key	City
Key	Zip
Key	Phone

Order	
PK	Order_number
Key	Customer_ID
Key	Customer_name
Key	To_street
Key	To_city
Key	To-state
Key	To-zip
Key	Ship_date

Product	
Key	Product_ID
Key	Quantity
Key	Product_type

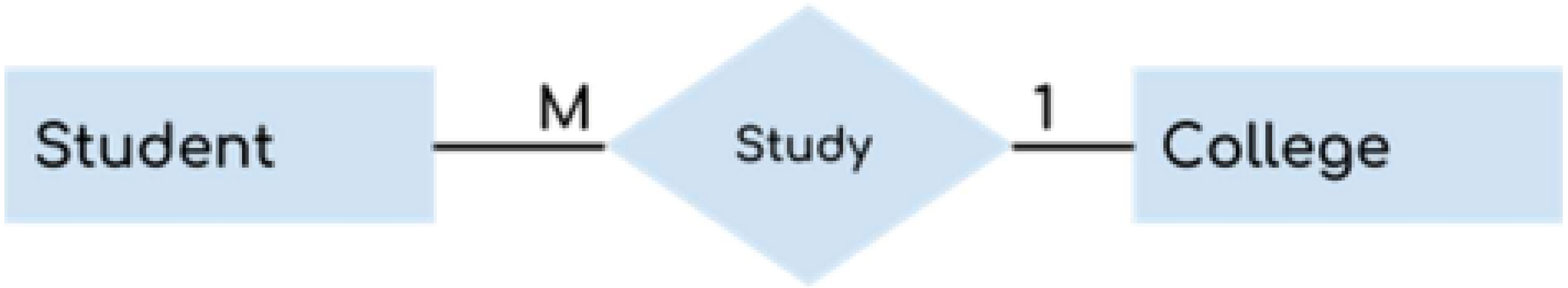
Relationships

Describe how entities interact with each other.



Identifying Entities

- An entity is an object or component of data. An entity is represented as rectangle in an ER diagram.
- The types of information that are saved in the database are called ***'entities'.***
- These entities exist in four kinds: people, things, events, and locations.



Example

- In the following ER diagram we have two entities Student and College and these two entities have many to one relationship as many students study in a single college.

Identifying Entities: An Example

- Creating a website for a shop. What are the possible entities?
- You sell your products to Customers
- The "Shop" is a location;
- "Sale" is an event;
- "Products" are things; and
- "Customers" are people.

Weak Entity

- An entity that cannot be uniquely identified by its own attributes and relies on the relationship with other entity is called weak entity.
- The weak entity is represented by a double rectangle.
- A weak entity is a type of entity which doesn't have its key attribute. It can be identified uniquely by considering the primary key of another entity.
- ***Weak entity sets need to have participation.***

Weak Entity

- A weak entity is a type of entity which doesn't have its key attribute. It can be identified uniquely by considering the primary key of another entity.

Examples of Weak Entities

Example 1:

- The existence of rooms is entirely dependent on the existence of a hotel. So room can be seen as the weak entity of the hotel.

Example 2:

The bank account of a particular bank has no existence if the bank doesn’t exist anymore.

Examples of Weak Entities

Example 3:

A company may store the information of dependents (Parents, Children, Spouse) of an Employee.

But the dependents don’t have existence without the employee. So Dependent will be weak entity type and Employee will be Identifying Entity type for Dependent.

Examples of Weak Entities

Example 4:

Product and Product Reviews: In an e-commerce database, 'Product Reviews' can be seen as a weak entity.

A 'Product Review' is inherently tied to the 'Product' it reviews - it cannot exist without the product.

Furthermore, the review's distinctive identifier often includes the product's ID, indicating its uniqueness dependence.

Why is it important to identify a weak entity in the ERD process?

1. Ensures Proper Representation of Real-World Relationships

Some entities in the real world (like **Dependents**, **Order Items**, **Room Bookings**, etc.) **cannot exist independently**.

2. **Helps Maintain Referential Integrity**

- Weak entities rely on **strong entities for identification**.
- By clearly marking them as weak, you enforce **referential integrity** — meaning a weak entity **must be linked** to an existing strong entity.

Why is it important to identify a weak entity in the ERD process?

3. **Guides the Creation of Correct Keys**

- Weak entities do **not have standalone primary keys**.
- Recognizing a weak entity ensures that its **primary key includes a foreign key** from its related strong entity, forming a **composite key**.

Why is it important to identify a weak entity in the ERD process?

4. Aids in Database Normalization and Design Efficiency

- Proper identification prevents redundancy and **data anomalies**.
- It clarifies entity roles, ensuring **normalized tables** with **minimal duplication** of data.

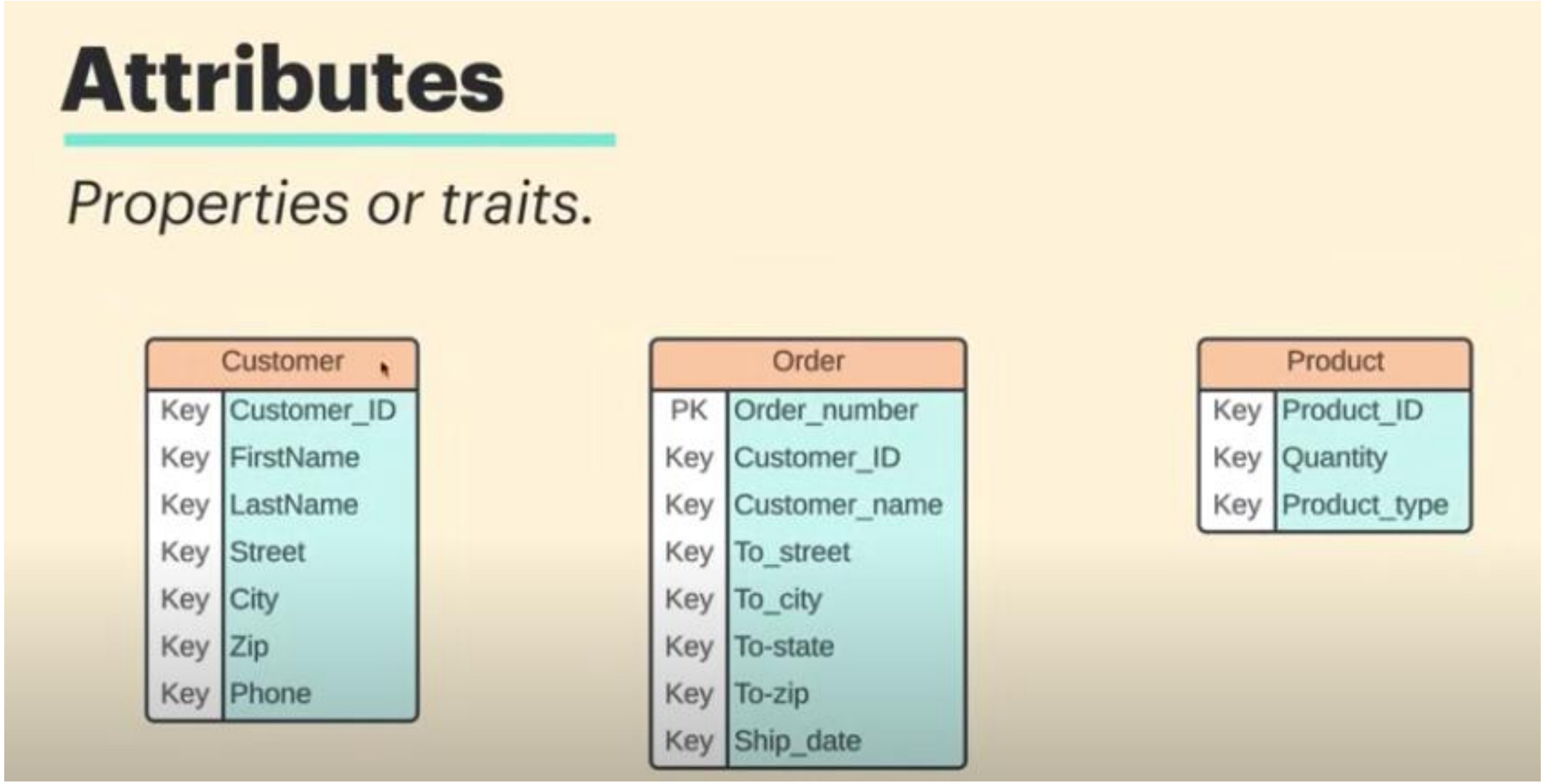
5. **Improves Clarity for Developers and Stakeholders**

- ERDs serve as blueprints. Labeling weak entities makes **relationships and constraints explicit**, helping teams understand **how data connects and depends** across the system.

Strong Entity Set vs. Weak Entity Set

Strong Entity Set	Weak Entity Set
Strong entity set always has a primary key.	It does not have enough attributes to build a primary key.
It is represented by a rectangle symbol.	It is represented by a double rectangle symbol.
It contains a Primary key represented by the underline symbol.	It contains a Partial Key which is represented by a dashed underline symbol.
The member of a strong entity set is called as dominant entity set.	The member of a weak entity set called as a subordinate entity set.
Primary Key is one of its attributes which helps to identify its member.	In a weak entity set, it is a combination of primary key and partial key of the strong entity set.
In the ER diagram the relationship between two strong entity set shown by using a diamond symbol.	The relationship between one strong and a weak entity set shown by using the double diamond symbol.
The connecting line of the strong entity set with the relationship is single.	The line connecting the weak entity set for identifying relationship is double.

Attributes

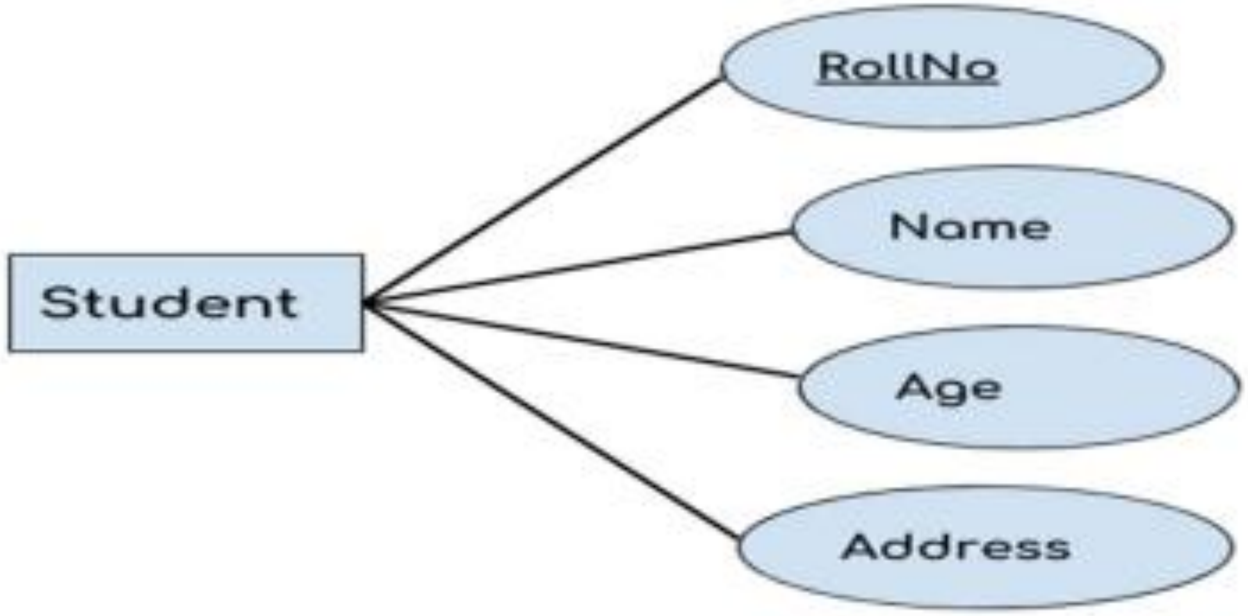

Attributes
Properties or traits.
Customer
Key Customer_ID
Key FirstName
Key LastName
Key Street
Key City
Key Zip
Key Phone
Order
PK Order_number
Key Customer_ID
Key Customer_name
Key To_street
Key To_city
Key To-state
Key To-zip
Key Ship_date
Product
Key Product_ID
Key Quantity
Key Product_type

Attributes

- An attribute describes the property of an entity.
- An attribute is represented as Oval in an ER diagram.
- There are four types of attributes:
    1. Key attribute
    2. Composite attribute
    3. Multivalued attribute
    4. Derived attribute

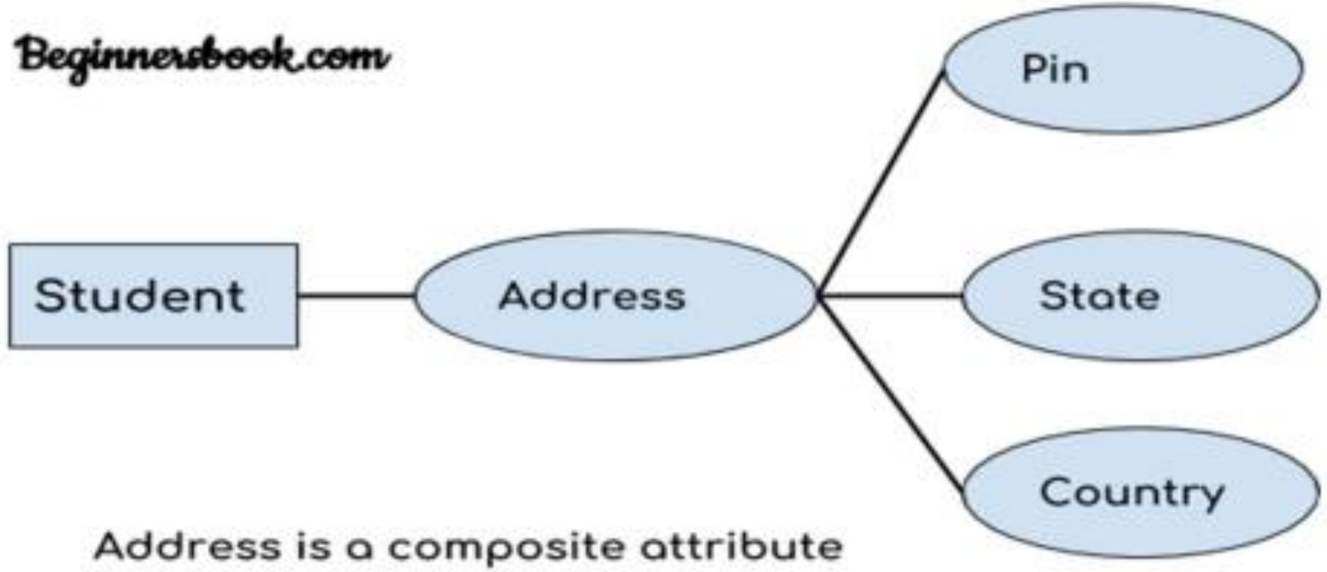
Key Attribute

- 1. A key attribute can uniquely identify an entity from an entity set.
- For example, student roll number can uniquely identify a student from a set of students.
- Key attribute is represented by oval same as other attributes however the text of key attribute is underlined.



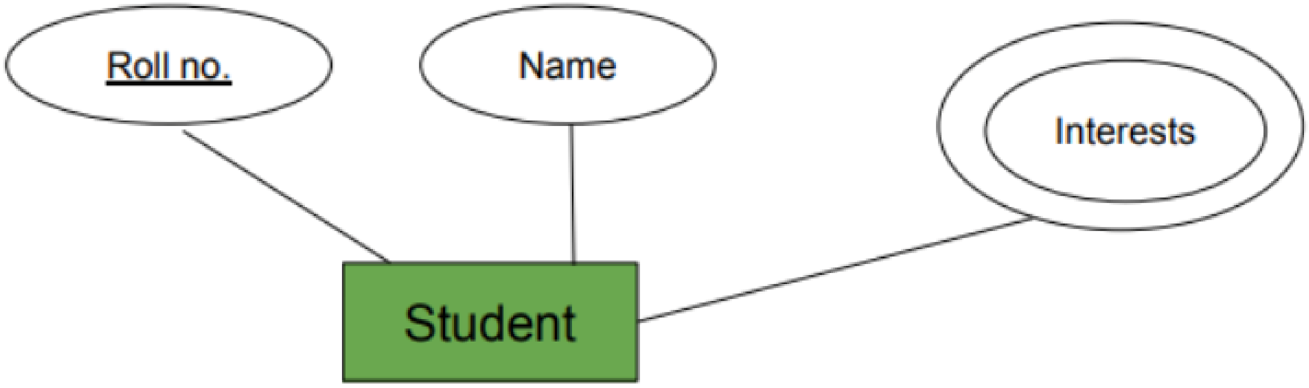
Composite Attribute

- An attribute that is a combination of other attributes is known as composite attribute.
- For example, in student entity, the student address is a composite attribute as an address is composed of other attributes such as pin code, state, country.

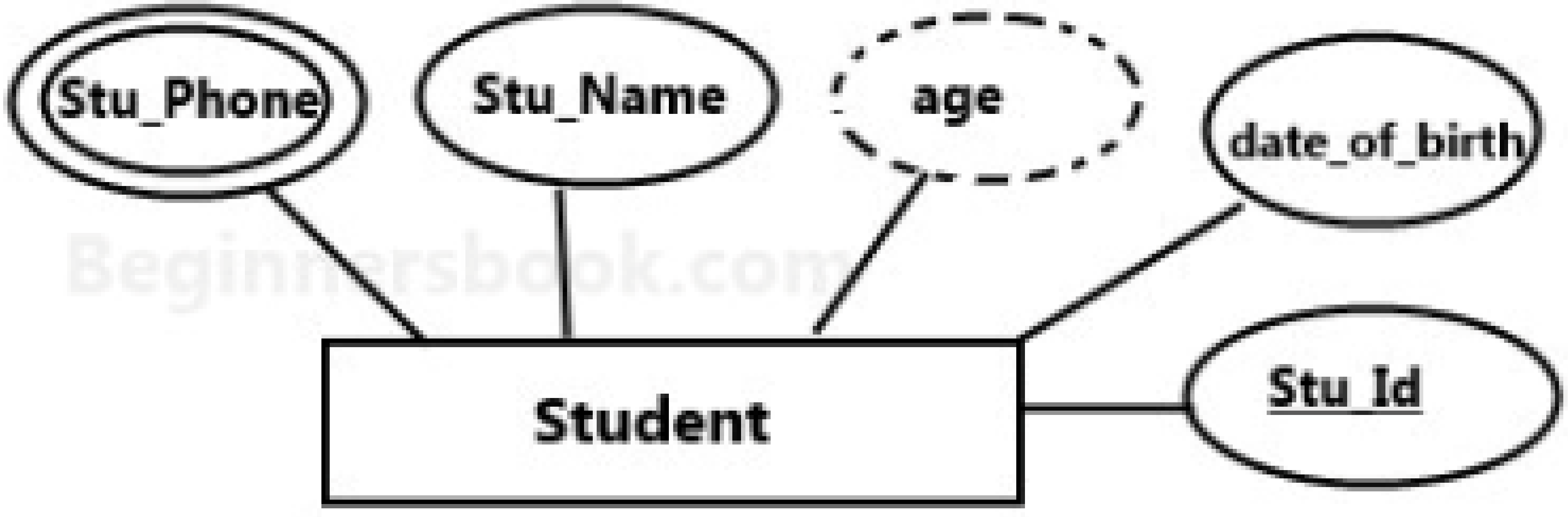


Multivalued Attribute

- A multivalued attribute allows more than one value to be stored for a single entity instance. In simpler words, it permits an entity to have several values assigned to a given attribute.
- **Example:** A student can have multiple phone numbers, or a company can have multiple divisions. Another example could be a person's skills, where they might have multiple skills listed (e.g., coding, writing, data analysis).



Student table with multivalued attribute (Interests).



Derived Attribute

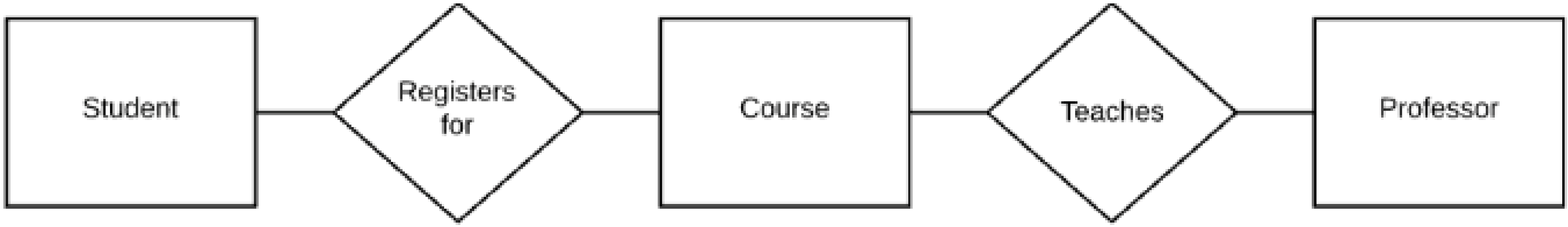
- A derived attribute is one whose value is dynamic and derived from another attribute.
- It is represented by dashed oval in an ER Diagram.
- For example – Person age is a derived attribute as it changes over time and can be derived from another attribute (Date of birth).

Relationships

- Describe how entities interact with each other.
- A relationship is represented by diamond shape in ER diagram, it shows the relationship among entities.
- There are four types of cardinal relationships:
  1. One to One
  2. One to Many
  3. Many to One
  4. Many to Many
- Cardinality: Defines the numerical attributes of the relationship between two entities or entity sets.

Identifying Entities: Identifying Relationships

- Customers --> Sales; 1 customer can buy something several times
- Sales --> Customers; 1 sale is always made by 1 customer at the time
- Customers --> Products; 1 customer can buy multiple products
- Products --> Customers; 1 product can be purchased by multiple customers
- Customers --> Shops; 1 customer can purchase in multiple shops
- Shops --> Customers, 1 shop can receive multiple customers
- Shops --> Products; in 1 shop there are multiple products
- Products --> Shops; 1 product (type) can be sold in multiple shops
- Shops --> Sales; in 1 shop multiple sales can me made
- Sales --> Shops; 1 sale can only be made in 1 shop at the time
- Products --> Sales; 1 product (type) can be purchased in multiple sales
- Sales --> Products; 1 sale can exist out of multiple products



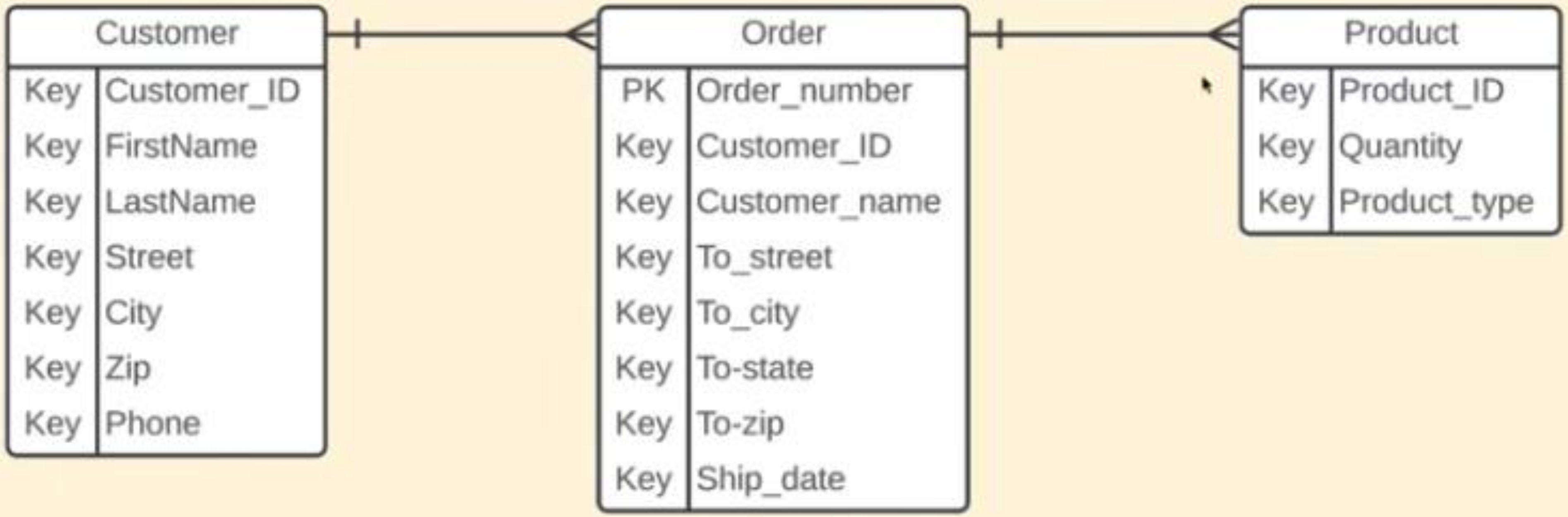
Identifying Relationships

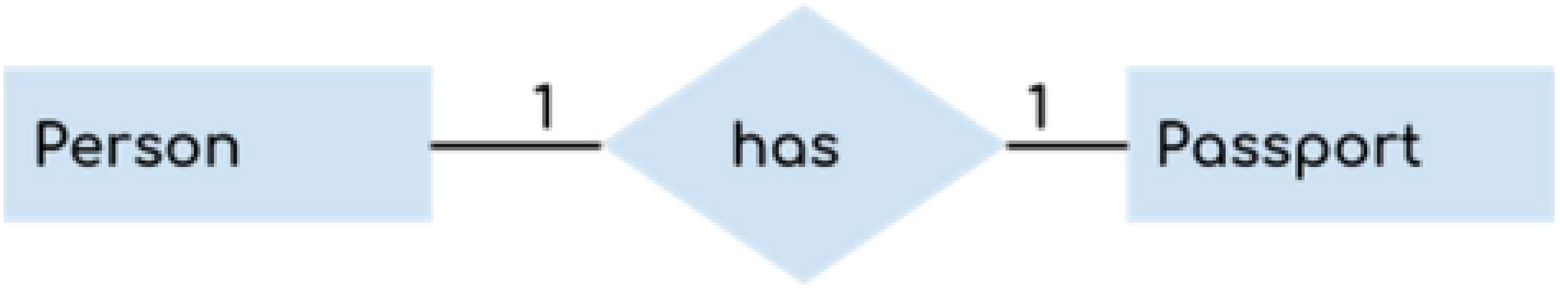
- **Relationships highlight how entities interact with each other.**

Relationships are typically verbs such as “buys,” “contains,” or “does.” In our example, the relationships “Registers for” and “Teaches” effectively explain the interactions between the three entities.

Relationships

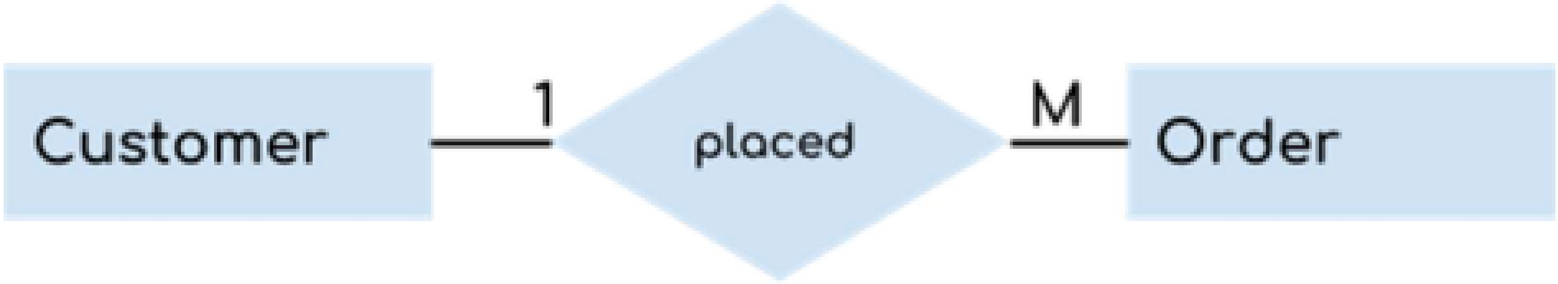
Describe how entities interact with each other.





One to One Relationship

- When a single instance of an entity is associated with a single instance of another entity then it is called one to one relationship.
- For example, a person has only one passport and a passport is given to one person.



One to Many Relationship

- When a single instance of an entity is associated with more than one instance of another entity, then it is called a one-to-many relationship.
- For example – a customer can place many orders, but an order cannot be placed by many customers.

Many to One Relationship

- When more than one instance of an entity is associated with a single instance of another entity, then it is called a many-to- one relationship.
- For example – many students can study in a single college but a student cannot study in many colleges at the same time.

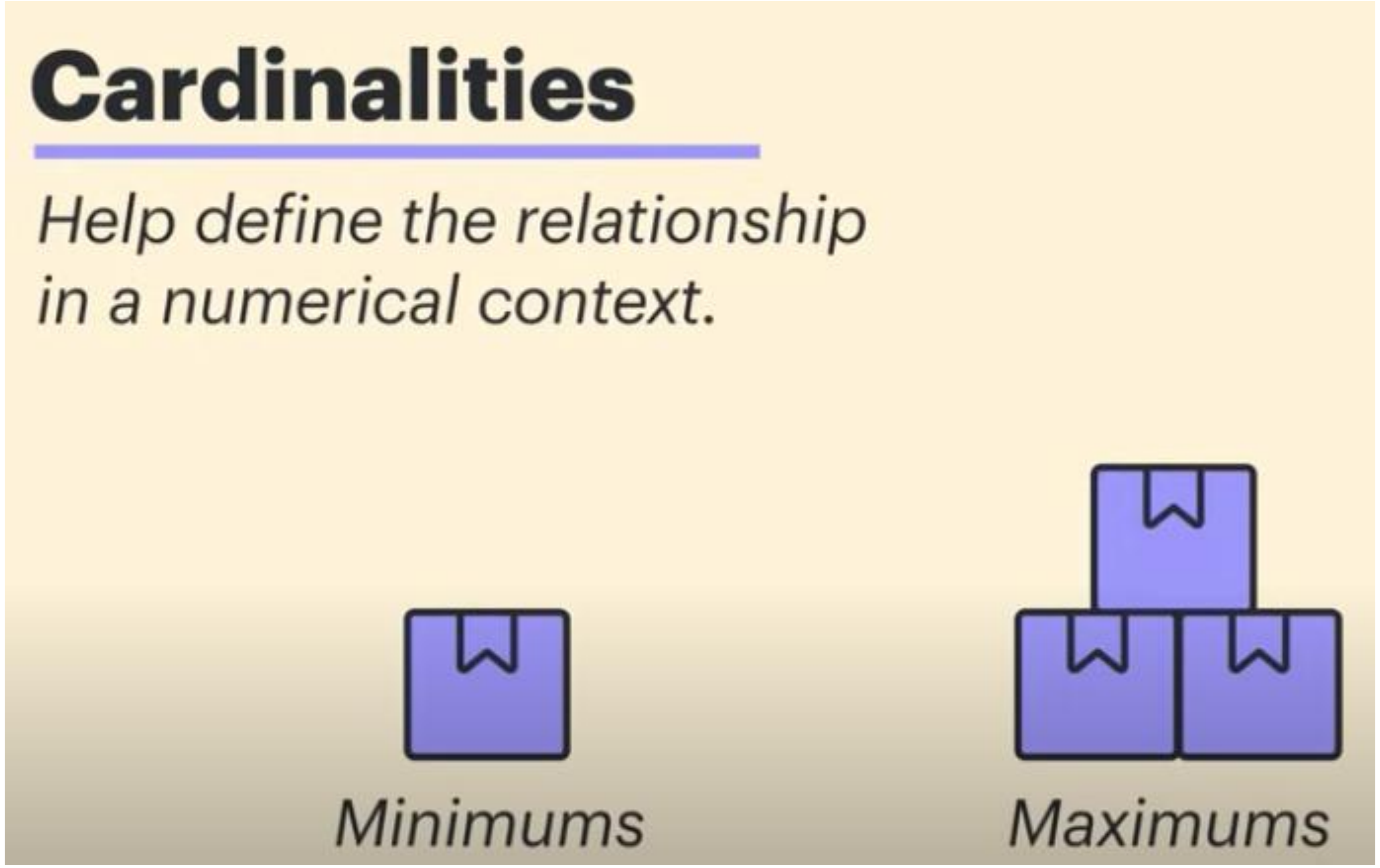
Many to Many Relationship

- When more than one instance of an entity is associated with more than one instance of another entity, then it is called a many-to- many relationship.
- For example, a can be assigned to many projects and a project can be assigned to many students.

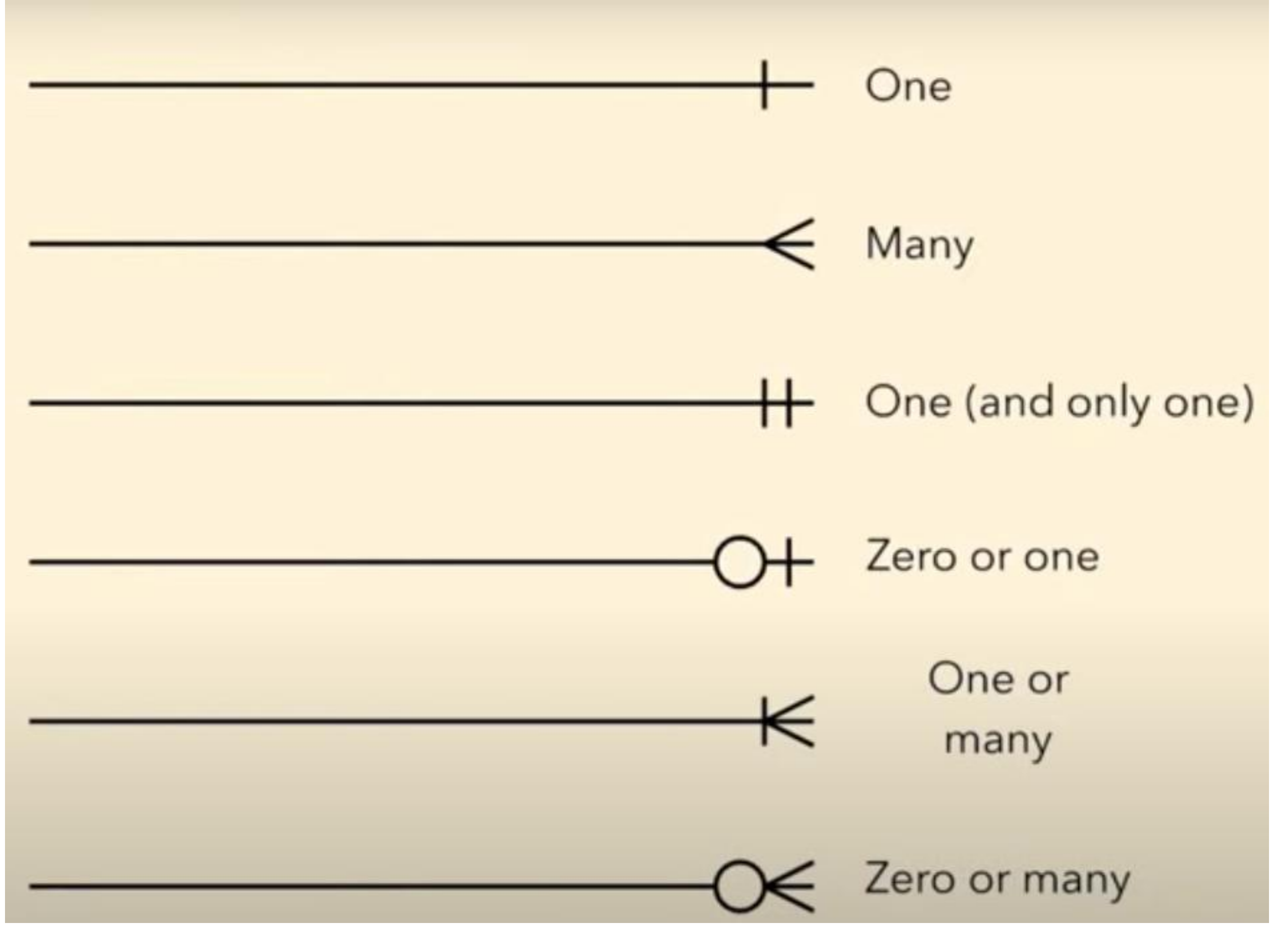
Relationships

- First, you need to state for each relationship, how much of one side belongs to exactly 1 of the other side.
- For example: How many customers belong to 1 sale?; How many sales belong to 1 customer?; How many sales take place in 1 shop?

Cardinalities



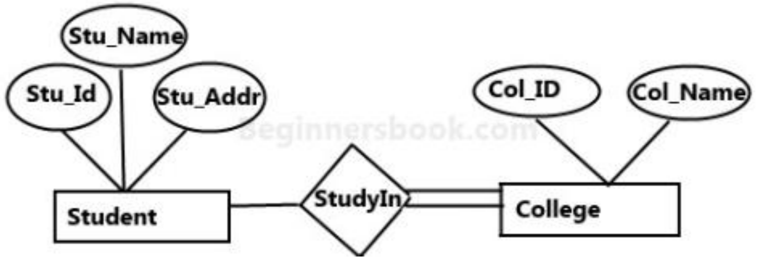
Cardinalities



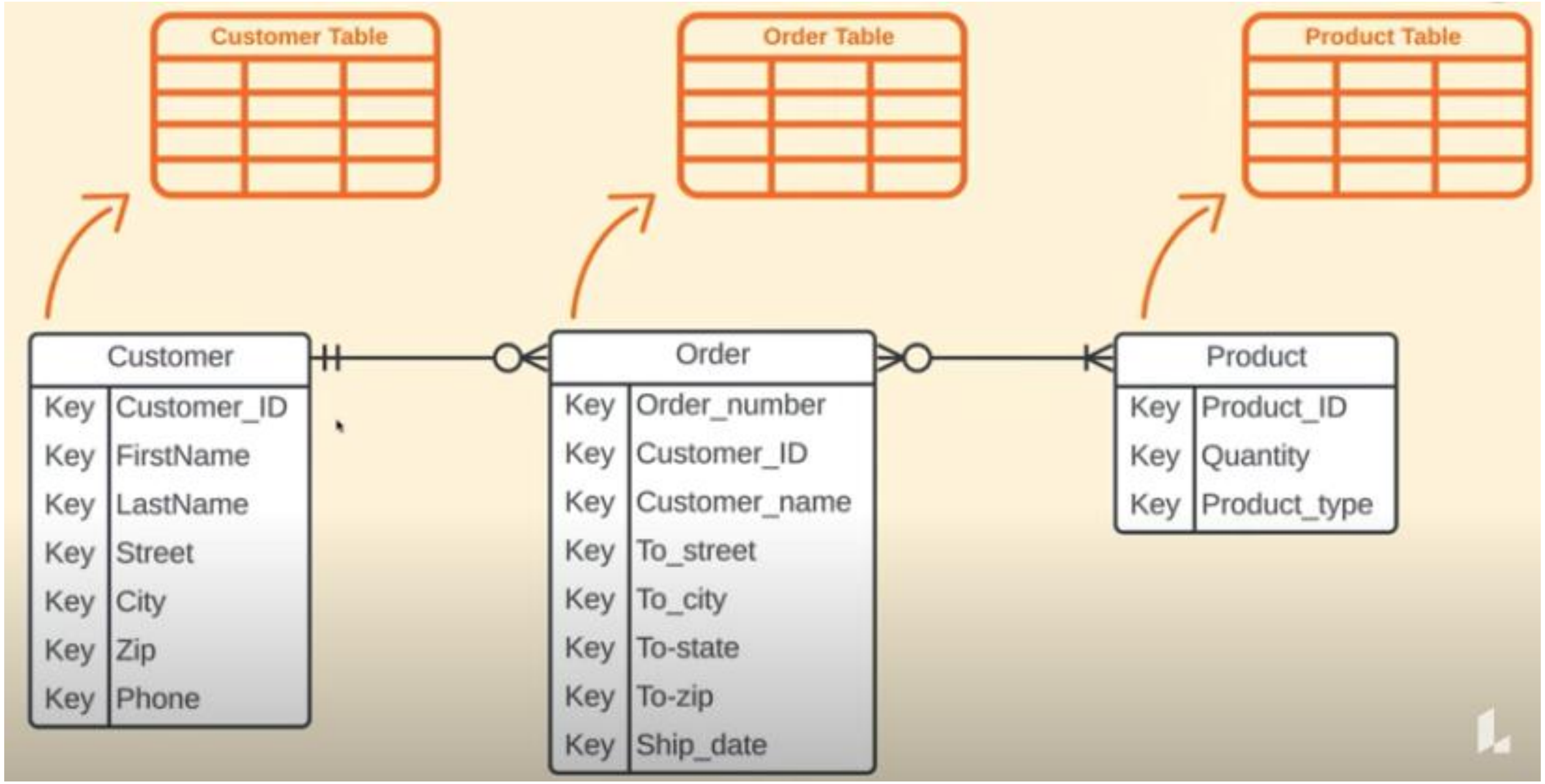
Total Participation of an Entity set

- A total participation of an entity set represents that each entity in an entity set must have at least one relationship in a relationship set.
- For example, In the diagram below, each college must have at least one associated Student.

E-R Digram with total participation of College entity set in StudyIn relationship Set - This indicates that each college must have atleast one associated Student.



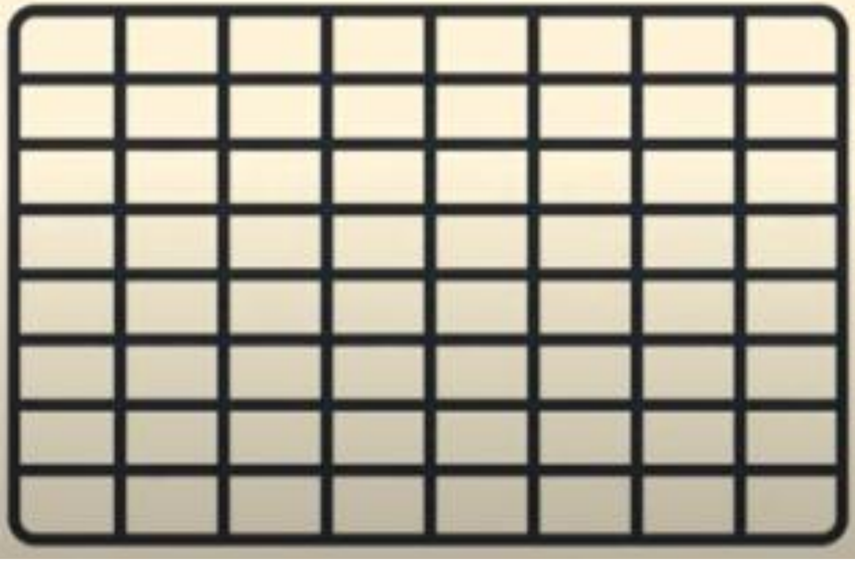
Each entity in the diagram represents a table



Primary Key

An attribute or field that uniquely identifies every record within a certain table.

Table



Primary Key Rules

1. A primary key has to be unique and identify with only one record in our table.
2. It needs to be never changing.
3. A primary key needs to be never null.

Sample Data

- What field uniquely identifies the records?

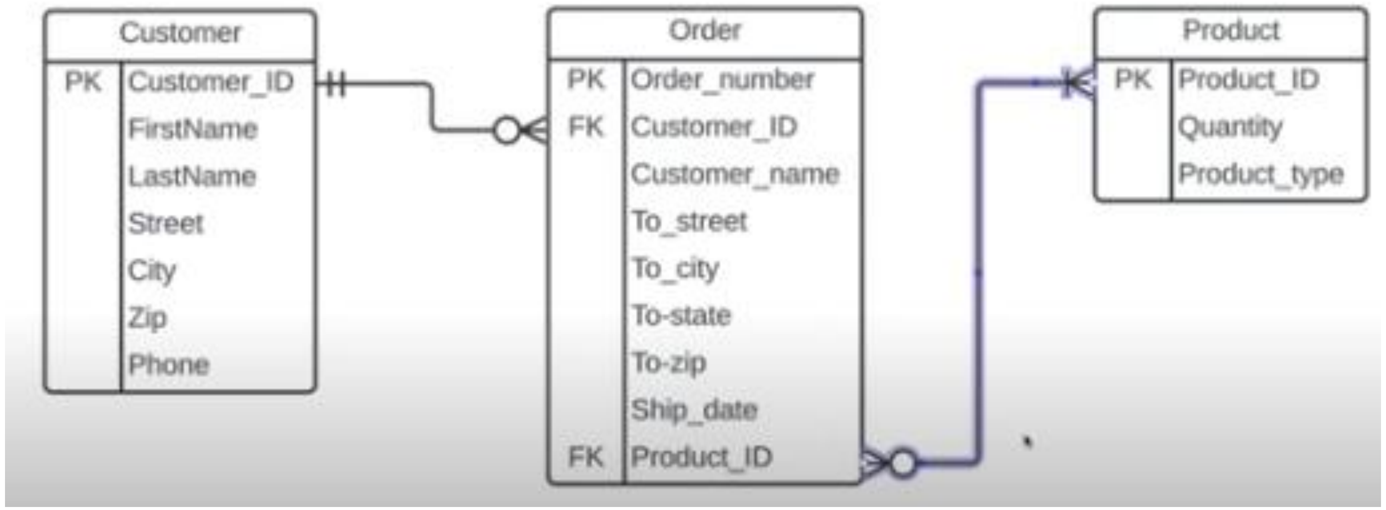
Customer_ID	FirstName	LastName	Street	City	Zip	Phone
30011	Linda	McGrath	7249 N. Bow Ridge St.	Ft Mitchell	53465	984-501-8146
30012	Iris	Edmunds	7135 North Rocky River Court	Yorktown	51280	142-194-7654
30013	Chandra	Parsons	847 Tanglewood Dr.	Calhoun	27265	457-946-0751
30014	Ranee	Peters	696 Fawn Court	Albany	77947	536-501-5470
30015	Steven	Langdon	64 Pennington Ave.	Jacksonville	87321	476-530-5632
30016	John	Smith	7411 Shirley Street	Springfield	87954	178-273-3949
30017	Ben	Chapman	6 James Ave.	Hopkinsville	74103	394-701-9589
30018	Jeremy	Nash	76 Strawberry Court	Billerica	31539	986-712-0139
30019	Rhett	Buckland	243 Mayflower St.	Watertown	37188	481-825-6516
30020	Carmen	Jones	8318 Mammoth Ave.	Lorton	25068	809-089-9715
30021	Marylynn	Smith	7411 Shirley Street	Springfield	87954	178-273-3949
30022	Dorothy	Taylor	845 South Bay Meadows Dr.	Trumbull	30499	521-864-3194
30023	Lena	Clarkson	50 Westport Rd.	Valparaiso	96270	576-225-2108
30024	Catheryn	Terry	921 Cardinal Court	Norwich	12826	807-981-9716
30025	Henry	Mackay	9927 Morris Ave.	Edison	96404	676-190-0191
30026	John	Smith	14 Lakewood Ave.	Centerville	88711	858-772-6359
30027	Irene	Ferguson	9100 N. Sleepy Hollow Street	Harlingen	59656	497-430-6649

Other examples of primary keys other than

- Usernames – in which some applications does not allow its users to change its username. If that is the case, the usernames are set as primary key to refer to the record of a certain person.

Foreign Keys

- Unlike a primary key, a foreign key does not have to be unique. It can be repeated inside a table.
- It is possible to have multiple foreign keys in one entity.



Composite Primary Key

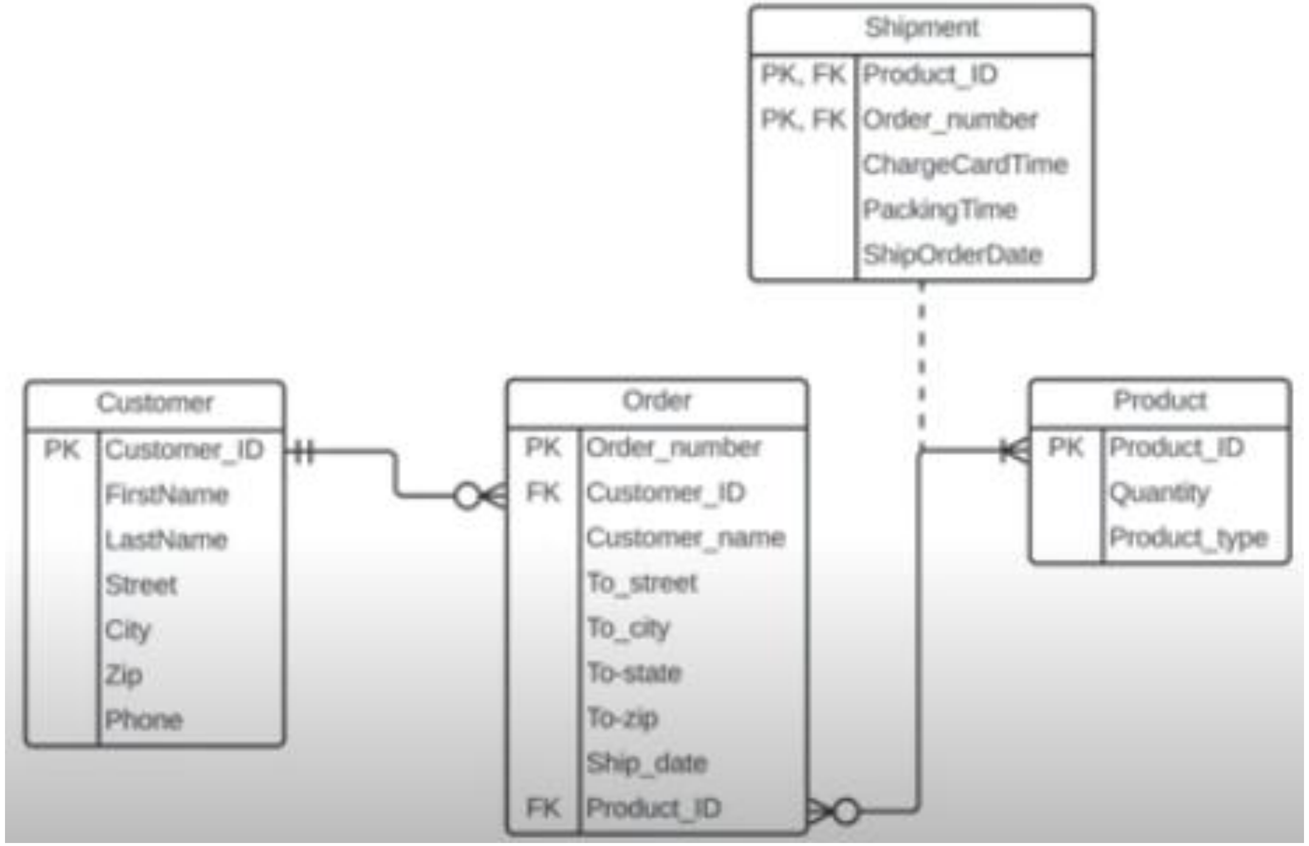
- Are used when two or more attributes are necessary to uniquely identify every record in a table.

1	Product_ID	Order_number	ChargeCardTime	PackingTime	ShipOrderDate
13	39792	252349915	6/1/2017 9:13:34	6/2/2017 10:14:46	6/3/2017 11:15:52
14	68010	252349916	6/1/2017 9:14:16	6/2/2017 10:15:02	6/3/2017 11:16:03
15	90884	252349917	6/1/2017 9:14:01	6/2/2017 10:15:26	6/3/2017 11:16:13
16	54486	252349918	6/1/2017 9:14:21	6/2/2017 10:15:39	6/3/2017 11:16:19
17	69246	252349919	6/1/2017 9:14:34	6/2/2017 10:15:41	6/3/2017 11:16:47
18	69253	252349919	6/1/2017 9:14:34	6/2/2017 10:15:45	6/3/2017 11:16:47
19	50319	252349920	6/1/2017 9:15:07	6/2/2017 10:16:07	6/3/2017 11:17:11
20	82552	252349921	6/1/2017 9:15:14	6/2/2017 10:16:07	6/3/2017 11:17:23
21	39172	252349922	6/1/2017 9:15:33	6/2/2017 10:16:28	6/3/2017 11:17:23
22	66997	252349923	6/1/2017 9:15:33	6/2/2017 10:16:50	6/3/2017 11:17:41
23	29257	252349924	6/1/2017 9:16:17	6/2/2017 10:17:05	6/3/2017 11:18:00
24	69253	252349925	6/1/2017 9:16:21	6/2/2017 10:17:17	6/3/2017 11:18:04
25	59640	252349926	6/1/2017 9:16:38	6/2/2017 10:17:29	6/3/2017 11:18:10
26	95813	252349927	6/1/2017 9:16:41	6/2/2017 10:17:30	6/3/2017 11:18:16
27	42101	252349928	6/1/2017 9:16:49	6/2/2017 10:17:44	6/3/2017 11:18:25
28	31229	252349929	6/1/2017 9:16:52	6/2/2017 10:17:53	6/3/2017 11:18:31
29	36382	252349930	6/1/2017 9:16:59	6/2/2017 10:17:58	6/2/2017 11:18:43

1	Product_ID	Order_number	ChargeCardTime	PackingTime	ShipOrderDate
13	39792	252349915	6/1/2017 9:13:34	6/2/2017 10:14:46	6/3/2017 11:15:52
14	68010	252349916	6/1/2017 9:14:16	6/2/2017 10:15:02	6/3/2017 11:16:03
15	90884	252349917	6/1/2017 9:14:01	6/2/2017 10:15:26	6/3/2017 11:16:13
16	54486	252349918	6/1/2017 9:14:21	6/2/2017 10:15:39	6/3/2017 11:16:19
17	69246	252349919	6/1/2017 9:14:34	6/2/2017 10:15:41	6/3/2017 11:16:47
18	69253	252349919	6/1/2017 9:14:34	6/2/2017 10:15:45	6/3/2017 11:16:47
19	50319	252349920	6/1/2017 9:15:07	6/2/2017 10:16:07	6/3/2017 11:17:11
20	82552	252349921	6/1/2017 9:15:14	6/2/2017 10:16:07	6/3/2017 11:17:23
21	39172	252349922	6/1/2017 9:15:33	6/2/2017 10:16:28	6/3/2017 11:17:23
22	66997	252349923	6/1/2017 9:15:33	6/2/2017 10:16:50	6/3/2017 11:17:41
23	29257	252349924	6/1/2017 9:16:17	6/2/2017 10:17:05	6/3/2017 11:18:00
24	69253	252349925	6/1/2017 9:16:21	6/2/2017 10:17:17	6/3/2017 11:18:04
25	59640	252349926	6/1/2017 9:16:38	6/2/2017 10:17:29	6/3/2017 11:18:10
26	95813	252349927	6/1/2017 9:16:41	6/2/2017 10:17:30	6/3/2017 11:18:16
27	42101	252349928	6/1/2017 9:16:49	6/2/2017 10:17:44	6/3/2017 11:18:25
28	31229	252349929	6/1/2017 9:16:52	6/2/2017 10:17:53	6/3/2017 11:18:31
29	36382	252349930	6/1/2017 9:16:59	6/2/2017 10:17:58	6/2/2017 11:18:43
30	81248	252349931	6/1/2017 9:17:02	6/2/2017 10:18:11	6/2/2017 11:18:54
31	61167	252349932	6/1/2017 9:17:07	6/2/2017 10:18:15	6/2/2017 11:19:04
32	75990	252349933	6/1/2017 9:17:14	6/2/2017 10:18:19	6/2/2017 11:19:16
33	83151	252349934	6/1/2017 9:17:18	6/2/2017 10:18:21	6/2/2017 11:19:18

Composite Primary Key Rules

- Use the fewest number of attributes as possible.
- Don’t use attributes that are apt to change.



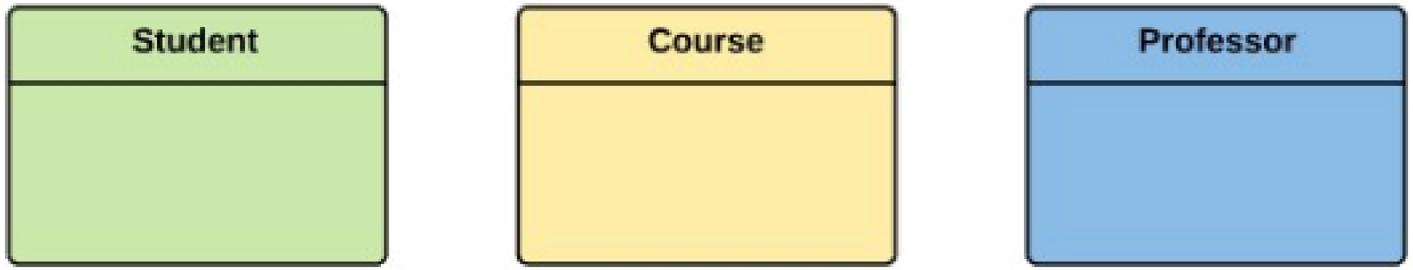
Steps to Create an ERD (E-R Digram)

Example

- In a university, a Student enrolls in Courses. A student must be assigned to at least one or more Courses. Each course is taught by a single Professor. To maintain instruction quality, a Professor can deliver only one course

Step 1) Entity Identification

- We have three entities
    - Student
    - Course
    - Professor



Step 2) Relationship Identification

- We have the following two relationships

 The student is assigned a course

 Professor delivers a course

Step 3) Cardinality Identification

- For them problem statement we know that,

 A student can be assigned multiple courses

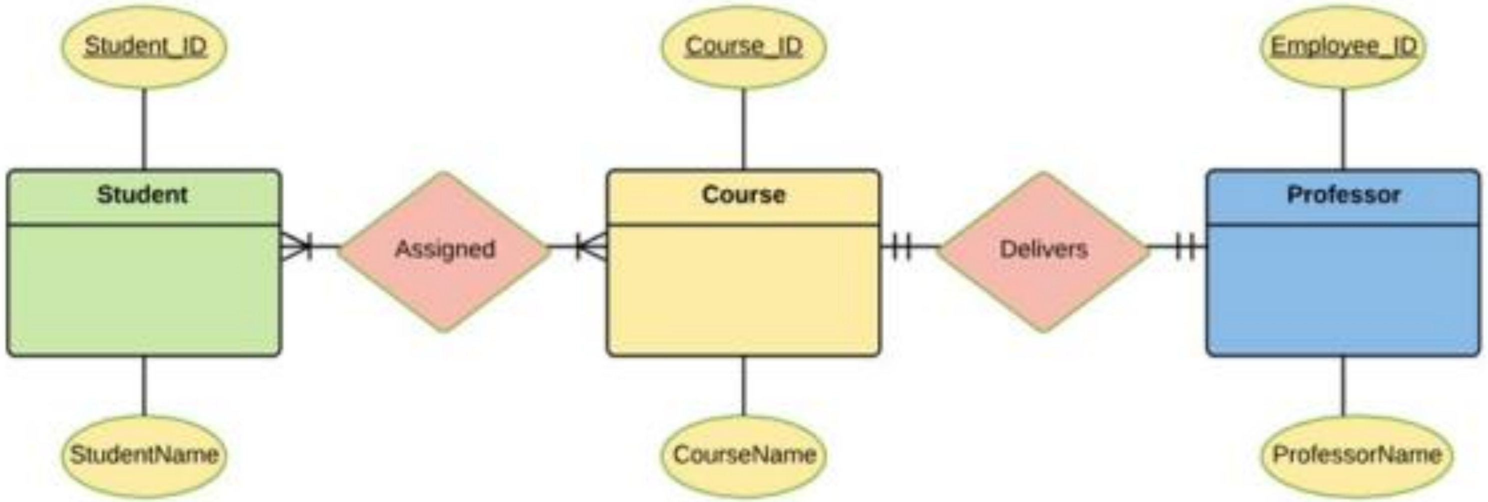
 A Professor can deliver only one course

Step 4) Identify Attributes

- You need to study the files, forms, reports, data currently maintained by the organization to identify attributes.
- You can also conduct interviews with various stakeholders to identify entities.
- Initially, it's important to identify the attributes without mapping them to a particular entity
- Once, you have a list of Attributes, you need to map them to the identified entities. Ensure an attribute is to be paired with exactly one entity.

Step 4) Identify Attributes

Entity	Primary Key	Attribute
Student	Student_ID	StudentName
Professor	Employee_ID	ProfessorName
Course	Course_ID	CourseName



Step 5) Create the ERD

- A more modern representation of ERD Diagram

Best Practices for Developing Effective ER Diagrams

- □ Eliminate any redundant entities or relationships
- □ You need to make sure that all your entities and relationships are properly labeled
- □ There may be various valid approaches to an ER diagram. You need to make sure that the ER diagram supports all the data you need to store
- □ You should ensure that each entity only appears a single time in the ER diagram
- □ Name every relationship, entity, and attribute represented on your diagram
- □ Never connect relationships to each other □ You should use colors to highlight important portions of the ER diagram

Summary

- The ER model is a high-level data model diagram
- ER diagrams are a visual tool which is helpful to represent the ER model
- Entity relationship diagram displays the relationships of entity set stored in a database
- ER diagrams help you to define terms related to entity relationship modeling
- ER model is based on three basic concepts: Entities, Attributes & Relationships
- An entity can be place, person, object, event or a concept, which stores data in the database